



**SIMATS**  
ENGINEERING



**SIMATS**  
Saveetha Institute of Medical And Technical Sciences  
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

### *Lab Programs*

#### **SET 26 - Student\_Mini\_Data.csv**

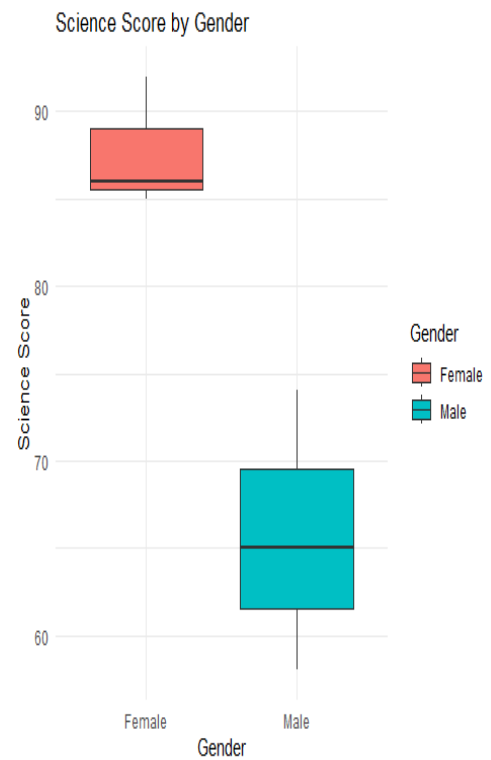
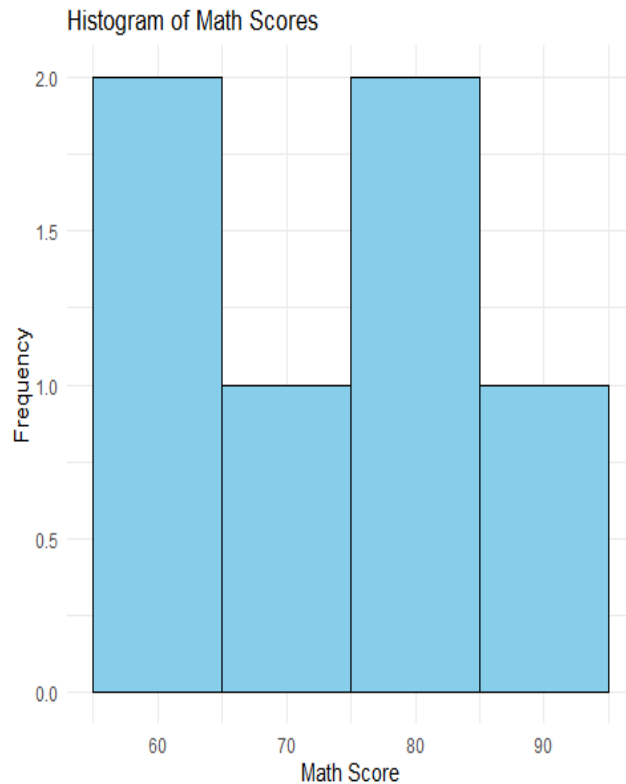
#### **26 A. Histogram & Boxplot**

##### **Program:**

```
student_data <- data.frame(  
  Student_ID=c("S01","S02","S03","S04","S05","S06"),  
  Gender=c("Male","Female","Male","Female","Male","Female"),  
  Study_Hours=c(2.0,3.5,1.5,4.0,2.8,3.0),  
  Attendance=c(78,90,70,95,85,92),  
  Math_Score=c(62,80,55,90,72,82),  
  Science_Score=c(65,85,58,92,74,86),  
  Exam_Date=c("2025-01-10","2025-01-10","2025-02-12",  
    "2025-02-12","2025-03-15","2025-03-15"))  
  
# Histogram  
ggplot(student_data,aes(x=Math_Score))+  
  geom_histogram(binwidth=10,fill="skyblue",color="black")+  
  labs(title="Histogram of Math Scores",  
    x="Math Score",  
    y="Frequency")+  
  theme_minimal()  
  
# Boxplot  
ggplot(student_data,  
  aes(x=Gender,y=Science_Score,fill=Gender))+  
  geom_boxplot()+  
  labs(title="Science Score by Gender",  
    x="Gender",  
    y="Science Score")+
```

```
theme_minimal()
```

### Output:

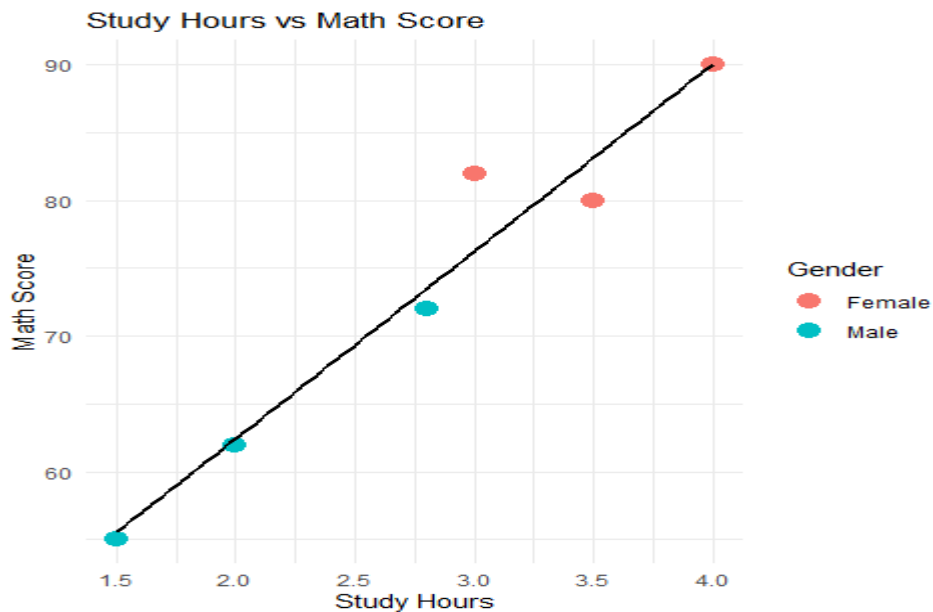


## 26 B. Scatter Plot with Regression Line

### Program:

```
ggplot(student_data,  
  aes(x=Study_Hours,  
    y=Math_Score,  
    color=Gender))+  
geom_point(size=4)+  
geom_smooth(method="lm",se=FALSE,color="black")+  
labs(title="Study Hours vs Math Score",  
  x="Study Hours",  
  y="Math Score")+  
theme_minimal()
```

## Output:



## 26 C. Monthly Average Math Scores with Moving Average

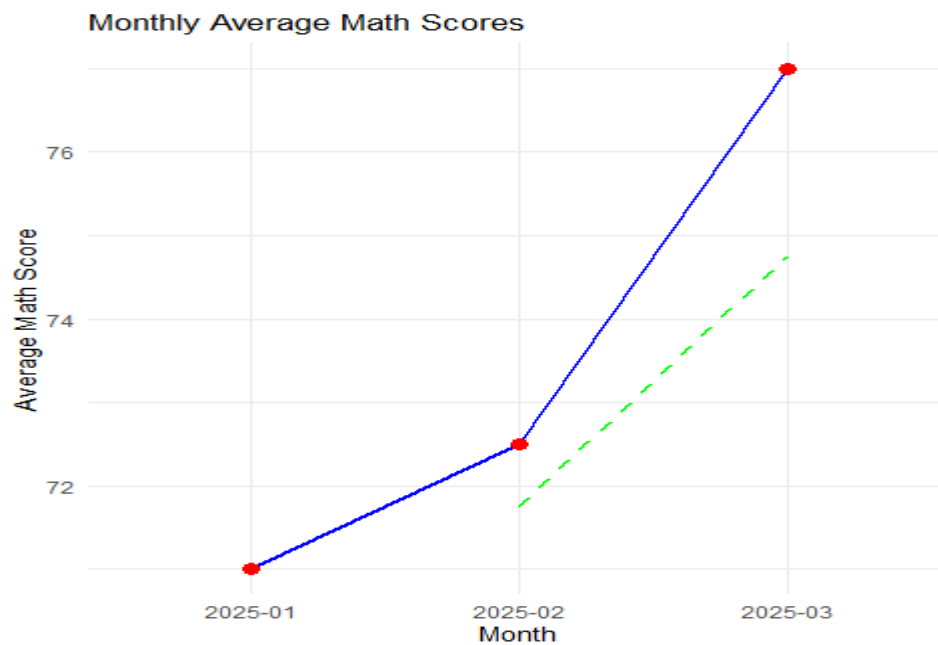
### Program:

```
student_data$Exam_Date <- as.Date(student_data$Exam_Date)
student_data$Month <- format(student_data$Exam_Date,"%Y-%m")
monthly_avg <- student_data %>%
  group_by(Month) %>%
  summarise(Average_Math=mean(Math_Score))
monthly_avg$Moving_Average <-
  rollmean(monthly_avg$Average_Math,
    k=2,
    fill=NA,
    align="right")

ggplot(monthly_avg,
  aes(x=Month,
    y=Average_Math,
    group=1))+
  geom_line(color="blue",linewidth=1)+
```

```
geom_point(size=3,color="red")+
geom_line(aes(y=Moving_Average),
          color="green",
          linewidth=1,
          linetype="dashed")+
labs(title="Monthly Average Math Scores",
      x="Month",
      y="Average Math Score")+
theme_minimal()
```

**Output:**



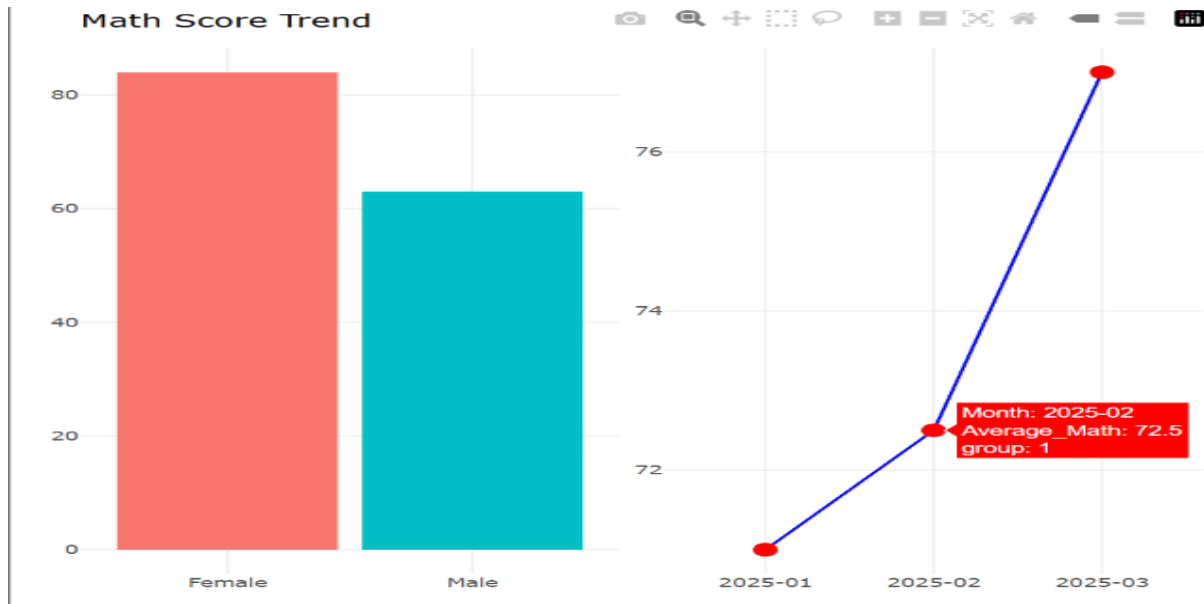
## 26 D. Interactive Dashboard (R Equivalent of Tableau)

**Program:**

```
gender_avg <- student_data %>%
  group_by(Gender) %>%
  summarise(Average_Math=mean(Math_Score))
bar_plot <- ggplotly(
  ggplot(gender_avg,
        aes(x=Gender,
            y=Average_Math,
            fill=Gender))+
```

```
geom_bar(stat="identity")+
labs(title="Average Math Score by Gender")+
theme_minimal()+
theme(legend.position="none"))
```

**Output:**



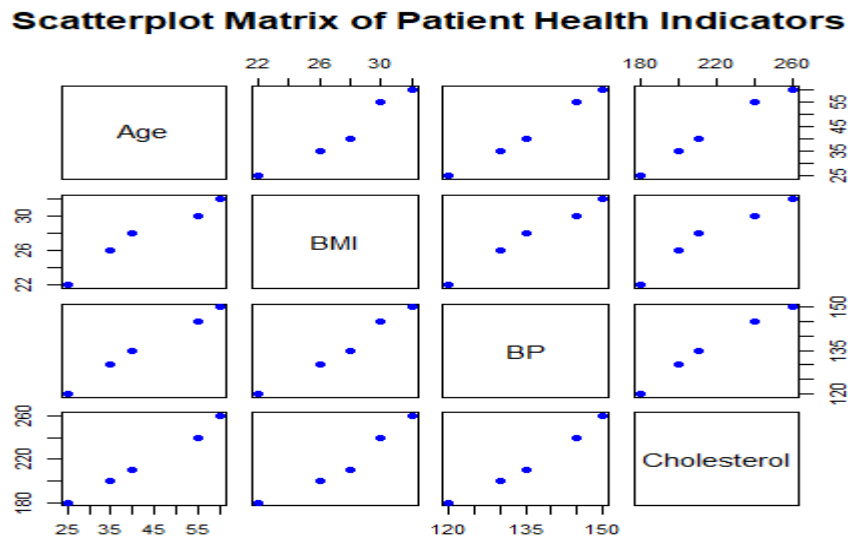
## SET 27 - Patient Health Risk Analysis

### 27 A. Scatterplot Matrix

**Program:**

```
patient_data <- data.frame(
  Patient_ID = c("P1","P2","P3","P4","P5"),
  Age = c(25,40,55,35,60),
  BMI = c(22,28,30,26,32),
  BP = c(120,135,145,130,150),
  Cholesterol = c(180,210,240,200,260))
pairs(
  patient_data[,2:5],
  main = "Scatterplot Matrix of Patient Health Indicators",
  pch = 19,
  col = "blue")
```

**Output:**



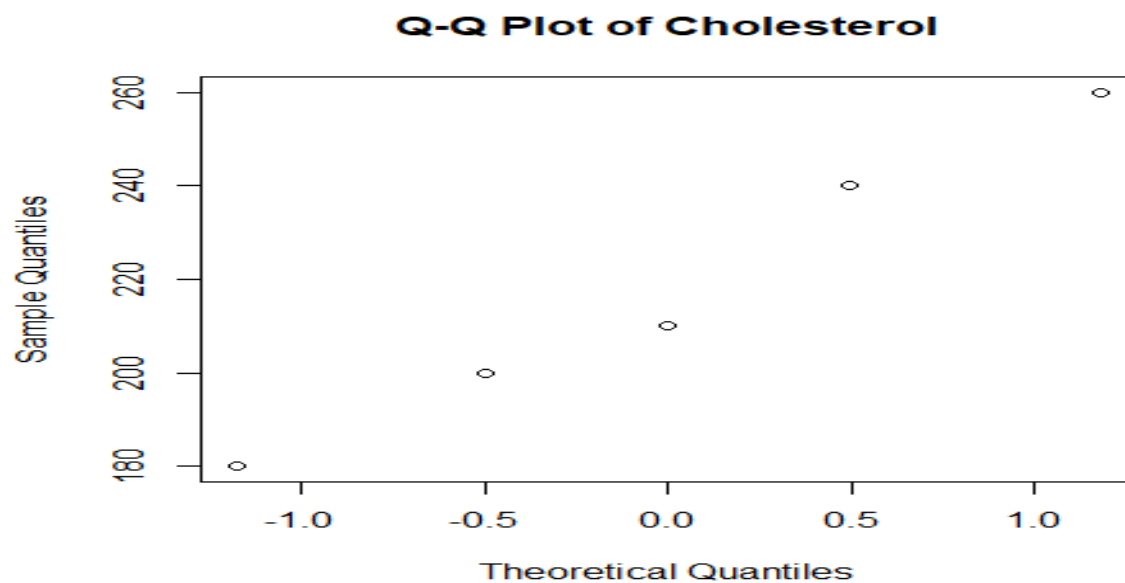
## 27 B. Q-Q Plot and ECDF

**Program:**

**Q-Q Plot:**

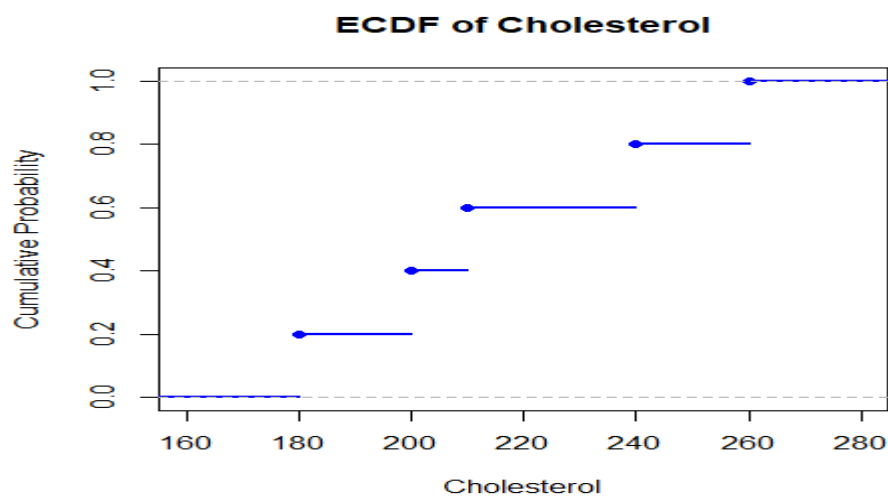
```
qqnorm(patient_data$Cholesterol,  
        main="Q-Q Plot of Cholesterol")  
qqline(patient_data$Cholesterol,  
        col="red")
```

**Output:**



**ECDF:**

```
plot(  
  ecdf(patient_data$Cholesterol),  
  main="ECDF of Cholesterol",  
  xlab="Cholesterol",  
  ylab="Cumulative Probability",  
  col="blue",  
  lwd=2)
```

**Output:****27 C. Bar Chart of Average Health Indicators****Program:**

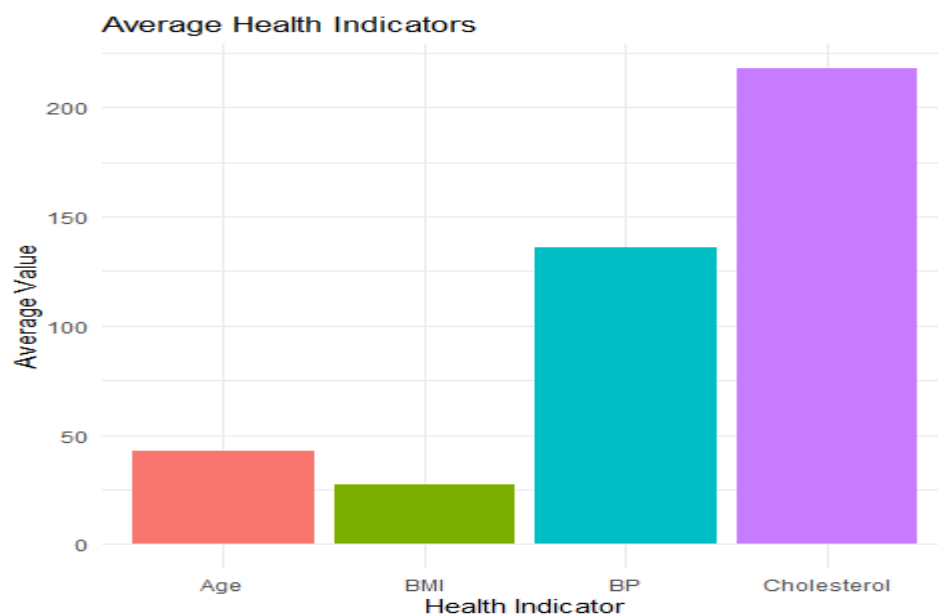
```
average_values <- data.frame(  
  Indicator = c("Age", "BMI", "BP", "Cholesterol"),  
  Average = c(  
    mean(patient_data$Age),  
    mean(patient_data$BMI),  
    mean(patient_data$BP),  
    mean(patient_data$Cholesterol)  
  ))  
ggplot(average_values,
```

```

aes(x=Indicator,
    y=Average,
    fill=Indicator))+
geom_bar(stat="identity")+
labs(title="Average Health Indicators",
     x="Health Indicator",
     y="Average Value")+
theme_minimal()+
theme(legend.position="none")

```

### Output:



## 27 D. Interactive Dashboard (R Equivalent of Tableau)

### Program:

```

patient_data$RiskScore <-
patient_data$BMI +
patient_data$BP +
patient_data$Cholesterol
heat_plot <- ggplotly(
  ggplot(patient_data,
    aes(x=Patient_ID,
        y=factor(Age),

```

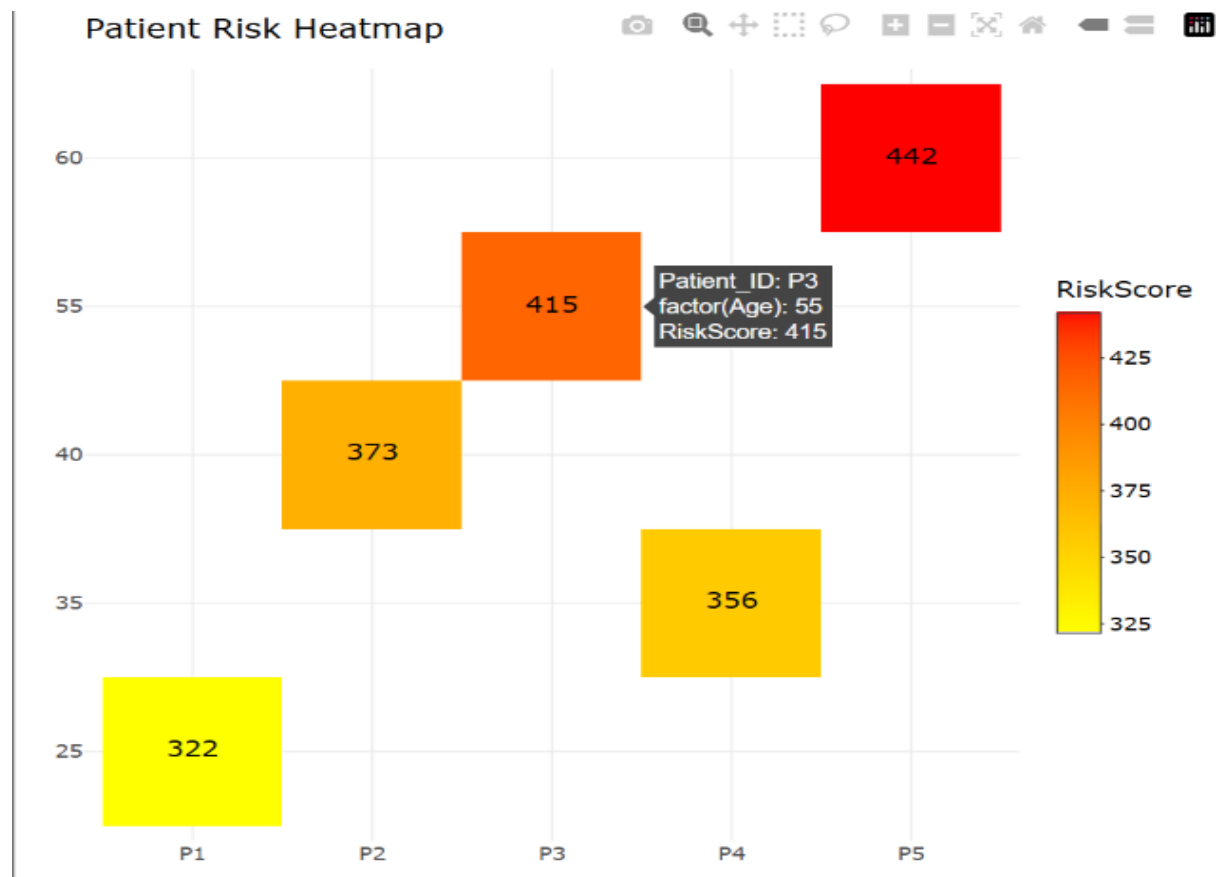


```

fill=RiskScore))+
geom_tile(color="white")+
geom_text(aes(label=RiskScore),
color="black")+
scale_fill_gradient(low="yellow",
high="red")+
labs(title="Patient Risk Heatmap",
x="Patient ID",
y="Age")+
theme_minimal()
dashboard <- subplot(
heat_plot,
nrows=1)
dashboard

```

### Output:



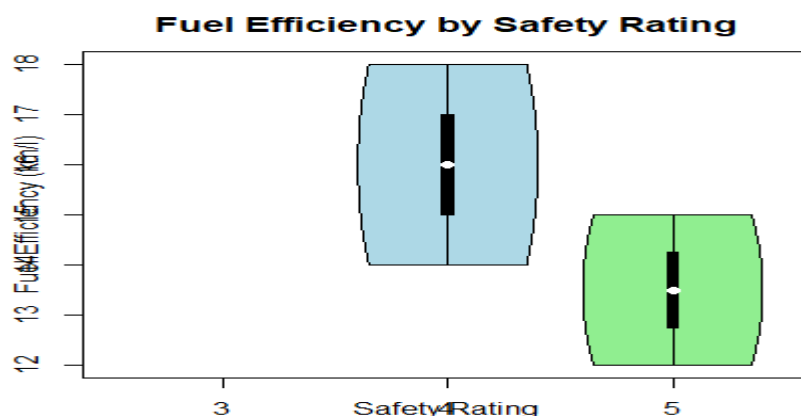
## SET 28. Vehicle Performance Analysis

### 28 A. violin plot

#### Program:

```
install.packages("vioplot")
library(vioplot)
vehicle <- data.frame(
  Vehicle_ID = c("V1", "V2", "V3", "V4", "V5"),
  Engine_Size = c(1.5, 2.0, 3.0, 2.5, 1.8),
  Horsepower = c(110, 150, 250, 200, 130),
  Fuel_Efficiency = c(18, 15, 12, 14, 17),
  Top_Speed = c(180, 200, 250, 220, 190),
  Safety_Rating = c(4, 5, 5, 4, 3))
vioplot(
  vehicle$Fuel_Efficiency[vehicle$Safety_Rating==3],
  vehicle$Fuel_Efficiency[vehicle$Safety_Rating==4],
  vehicle$Fuel_Efficiency[vehicle$Safety_Rating==5],
  names=c("3", "4", "5"),
  col=c("pink", "lightblue", "lightgreen"),
  main="Fuel Efficiency by Safety Rating",
  xlab="Safety Rating",
  ylab="Fuel Efficiency (km/l)")
```

#### Output:

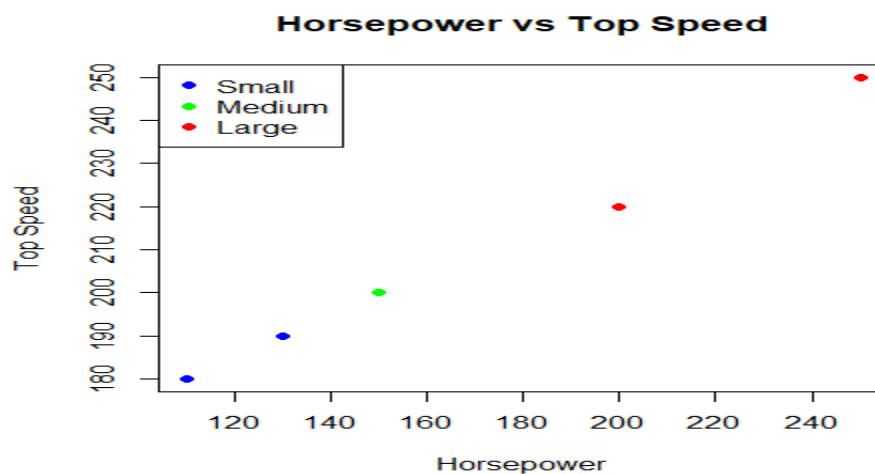


## 28 B. Scatter plot to explore the relationship between Horsepower and Top\_Speed

### Program:

```
vehicle <- data.frame(  
  Vehicle_ID = c("V1","V2","V3","V4","V5"),  
  Engine_Size = c(1.5,2.0,3.0,2.5,1.8),  
  Horsepower = c(110,150,250,200,130),  
  Fuel_Efficiency = c(18,15,12,14,17),  
  Top_Speed = c(180,200,250,220,190),  
  Safety_Rating = c(4,5,5,4,3))  
color <- ifelse(vehicle$Engine_Size < 2,"blue",  
  ifelse(vehicle$Engine_Size < 2.5,"green","red"))  
plot(vehicle$Horsepower,  
  vehicle$Top_Speed,  
  pch=19, col=color,  
  main="Horsepower vs Top Speed",  
  xlab="Horsepower",  
  ylab="Top Speed")  
legend("topleft",  
  legend=c("Small","Medium","Large"),  
  col=c("blue","green","red"),  
  pch=19)
```

### Output:

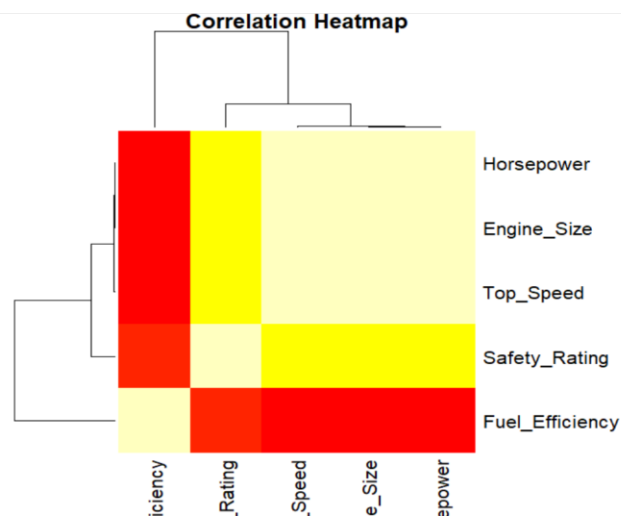


## 28 C. Correlation heatmap

### Program:

```
vehicle <- data.frame(  
  Vehicle_ID = c("V1","V2","V3","V4","V5"),  
  Engine_Size = c(1.5,2.0,3.0,2.5,1.8),  
  Horsepower = c(110,150,250,200,130),  
  Fuel_Efficiency = c(18,15,12,14,17),  
  Top_Speed = c(180,200,250,220,190),  
  Safety_Rating = c(4,5,5,4,3))  
cor_matrix <- cor(vehicle[,2:6])  
heatmap(cor_matrix,  
  col=heat.colors(10),  
  scale="none",  
  main="Correlation Heatmap")
```

### Output:



## 28 D. Interactive dashboard

### Program:

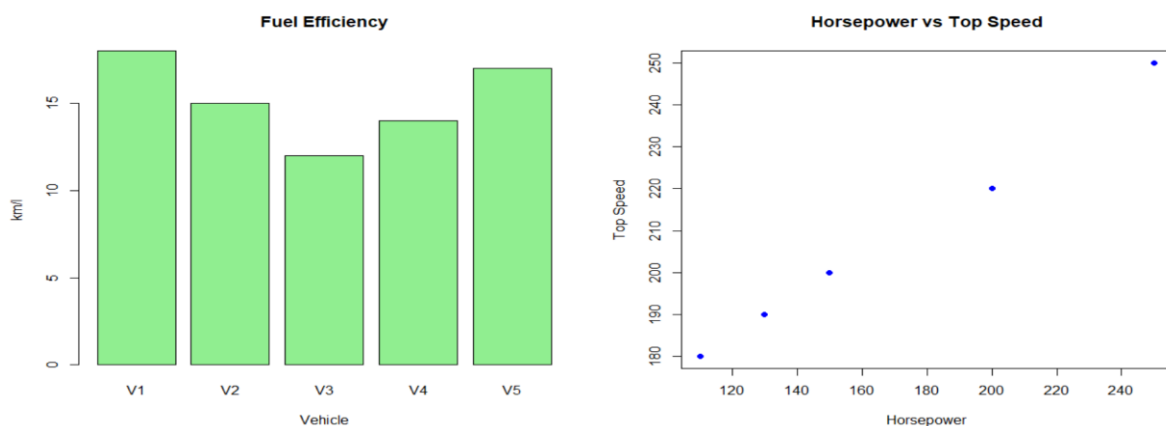
```
Vehicle_ID = c("V1","V2","V3","V4","V5"),  
Engine_Size = c(1.5,2.0,3.0,2.5,1.8),  
Horsepower = c(110,150,250,200,130),  
Fuel_Efficiency = c(18,15,12,14,17),
```

```

Top_Speed = c(180,200,250,220,190),
Safety_Rating = c(4,5,5,4,3))
par(mfrow=c(1,2))
barplot(vehicle$Fuel_Efficiency,
        names.arg=vehicle$Vehicle_ID,
        col="lightgreen",
        main="Fuel Efficiency",
        xlab="Vehicle",
        ylab="km/l")
plot(vehicle$Horsepower,
      vehicle$Top_Speed,
      pch=19,
      col="blue",
      main="Horsepower vs Top Speed",
      xlab="Horsepower",
      ylab="Top Speed")
par(mfrow=c(1,1))

```

### Output:



## SET 29 Student Academic Performance

### 29 A. Stacked area chart

#### Program:

```

student <- data.frame(
  Student_ID = c("S1","S2","S3","S4","S5"),

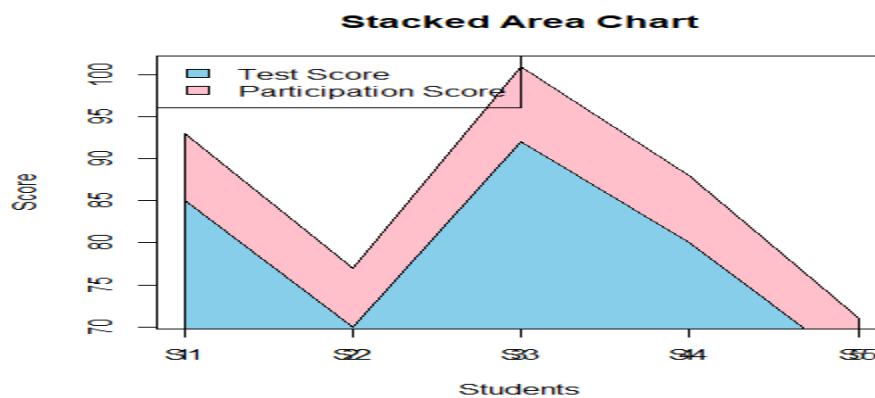
```

```

Age = c(19,21,20,22,23),
Study_Hours = c(12,8,15,10,7),
Attendance = c(90,70,95,85,60),
Test_Score = c(85,70,92,80,65),
Participation_Score = c(8,7,9,8,6))
x <- 1:5
y1 <- student$Test_Score
y2 <- student$Participation_Score
plot(x, y1+y2, type="n",
     xlab="Students",
     ylab="Score",
     main="Stacked Area Chart")
polygon(c(x,rev(x)),
       c(rep(0,5),rev(y1)),
       col="skyblue")
polygon(c(x,rev(x)),
       c(y1,rev(y1+y2)),
       col="pink")
axis(1,at=x,labels=student$Student_ID)
legend("topleft",
      legend=c("Test Score","Participation Score"),
      fill=c("skyblue","pink"))

```

**Output:**

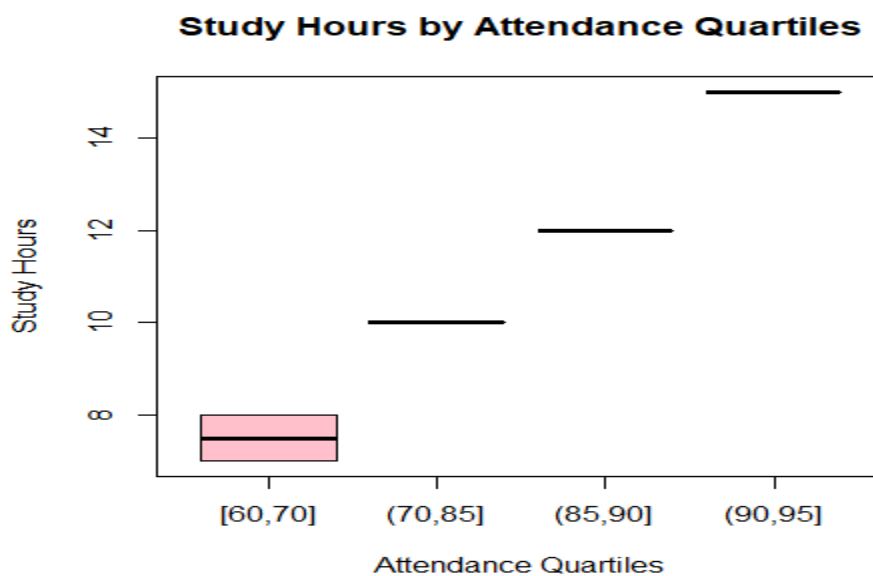


## 29 B. Boxplot of Study Hours grouped by Attendance quartiles

### Program:

```
student <- data.frame(  
  Student_ID = c("S1","S2","S3","S4","S5"),  
  Age = c(19,21,20,22,23),  
  Study_Hours = c(12,8,15,10,7),  
  Attendance = c(90,70,95,85,60),  
  Test_Score = c(85,70,92,80,65),  
  Participation_Score = c(8,7,9,8,6))  
student$Quartile <- cut(student$Attendance,  
  breaks=quantile(student$Attendance),  
  include.lowest=TRUE)  
boxplot(Study_Hours~Quartile,  
  data=student,  
  col=c("pink","lightgreen","skyblue","orange"),  
  main="Study Hours by Attendance Quartiles",  
  xlab="Attendance Quartiles",  
  ylab="Study Hours")
```

### Output:

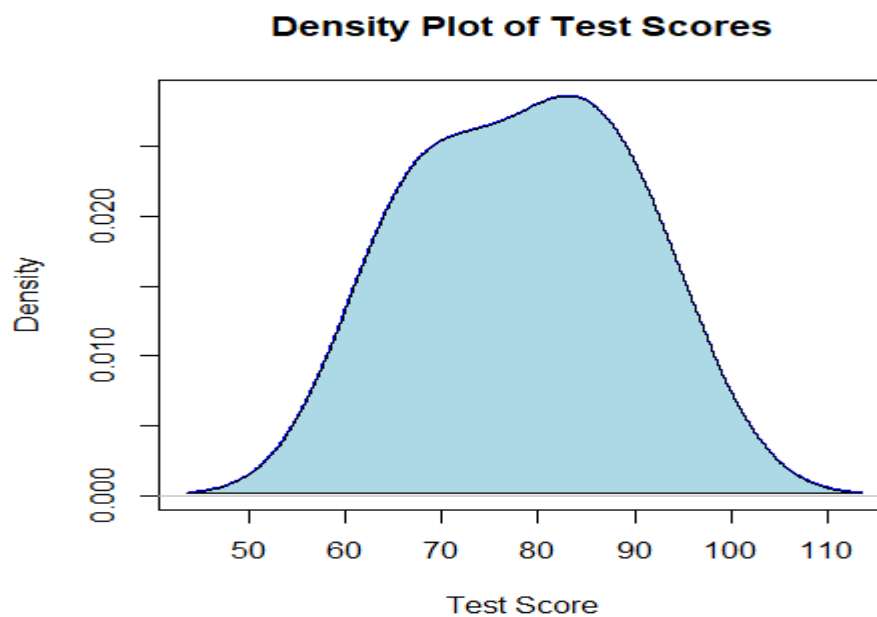


## 29 C. Density plot for Test\_score

### Program:

```
student <- data.frame(  
  Student_ID = c("S1","S2","S3","S4","S5"),  
  Age = c(19,21,20,22,23),  
  Study_Hours = c(12,8,15,10,7),  
  Attendance = c(90,70,95,85,60),  
  Test_Score = c(85,70,92,80,65),  
  Participation_Score = c(8,7,9,8,6))  
d <- density(student$Test_Score)  
plot(d,  
  col="blue",  
  lwd=2,  
  main="Density Plot of Test Scores",  
  xlab="Test Score",  
  ylab="Density")  
polygon(d,  
  col="lightblue")
```

### Output:



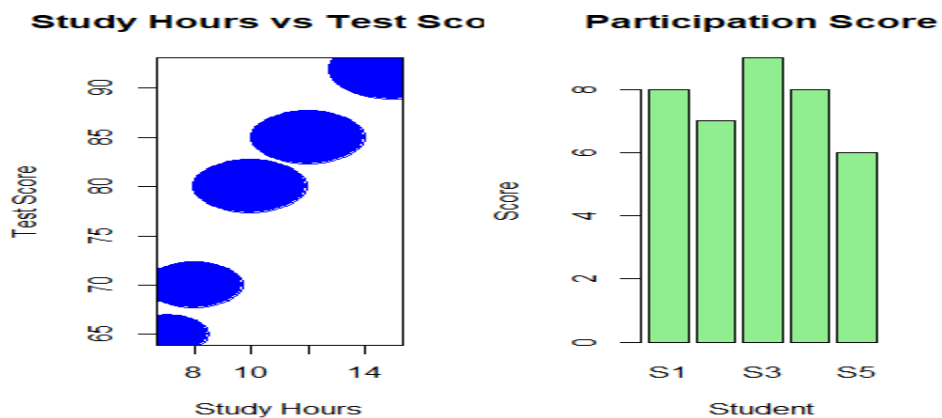


## 29 D. Interactive scatterplot with Study Hours

### Program:

```
student <- data.frame(  
  Student_ID = c("S1","S2","S3","S4","S5"),  
  Age = c(19,21,20,22,23),  
  Study_Hours = c(12,8,15,10,7),  
  Attendance = c(90,70,95,85,60),  
  Test_Score = c(85,70,92,80,65),  
  Participation_Score = c(8,7,9,8,6))  
par(mfrow=c(1,2))  
plot(student$Study_Hours,  
  student$Test_Score,  
  pch=19,col="blue",  
  cex=student$Participation_Score,  
  main="Study Hours vs Test Score",  
  xlab="Study Hours",  
  ylab="Test Score")  
barplot(student$Participation_Score,  
  names.arg=student$Student_ID,  
  xlab="Student",  
  ylab="Score")  
par(mfrow=c(1,1))
```

### Output:



## SET 30 Mobile App Usage

### 30 A. Histogram and density plot for Screen Time

#### Histogram for Screen Time

##### Program:

```
mobile <- data.frame(  
  User_ID = c("U01","U02","U03","U04","U05","U06"),  
  Gender = c("Male","Female","Male","Female","Male","Female"),  
  Age = c(20,22,19,21,23,20),  
  Screen_Time = c(4.5,6.0,3.2,7.1,2.8,5.4),  
  App_Usage_Count = c(18,25,12,30,10,22),  
  Data_Used = c(2.4,3.8,1.6,4.5,1.2,3.1),  
  Satisfaction = c(3,5,3,5,2,4),  
  Usage_Date = as.Date(c("2025-01-08","2025-01-08",  
    "2025-02-11","2025-02-11",  
    "2025-03-14","2025-03-14")))  
  
hist(mobile$Screen_Time,  
  col="skyblue",  
  main="Histogram of Screen Time",  
  xlab="Screen Time (hrs)",  
  ylab="Frequency")
```

##### Output:

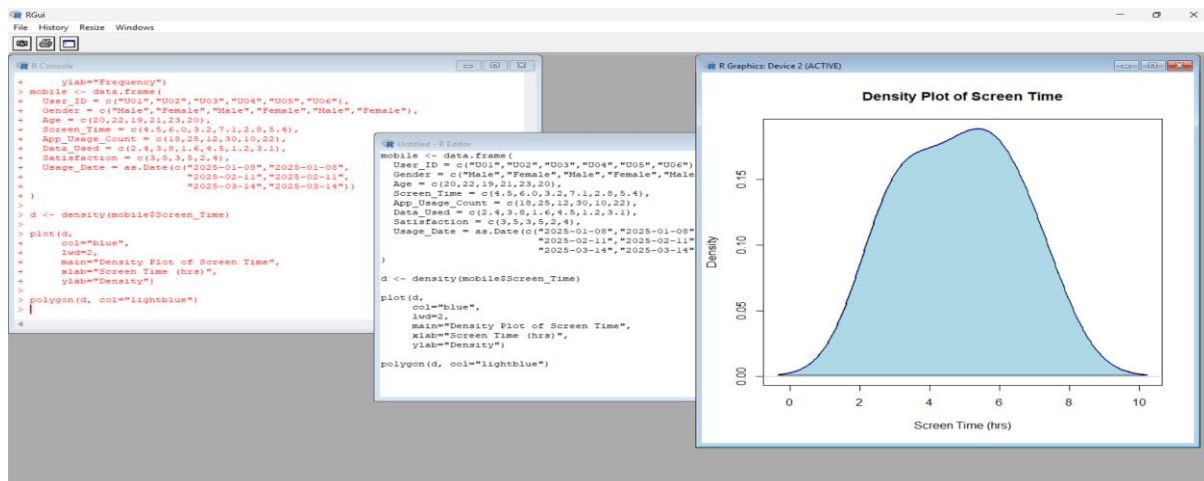


## Density Plot for Screen Time

### Program:

```
mobile <- data.frame(  
  User_ID = c("U01","U02","U03","U04","U05","U06"),  
  Gender = c("Male","Female","Male","Female","Male","Female"),  
  Age = c(20,22,19,21,23,20),  
  Screen_Time = c(4.5,6.0,3.2,7.1,2.8,5.4),  
  App_Usage_Count = c(18,25,12,30,10,22),  
  Data_Used = c(2.4,3.8,1.6,4.5,1.2,3.1),  
  Satisfaction = c(3,5,3,5,2,4),  
  Usage_Date = as.Date(c("2025-01-08","2025-01-08",  
    "2025-02-11","2025-02-11",  
    "2025-03-14","2025-03-14")))  
  
d <- density(mobile$Screen_Time)  
  
plot(d,  
  col="blue",  
  lwd=2,  
  main="Density Plot of Screen Time",  
  xlab="Screen Time (hrs)",  
  ylab="Density")  
  
polygon(d, col="lightblue")
```

### Output:



## 30 B. Scatter plot

### Program:

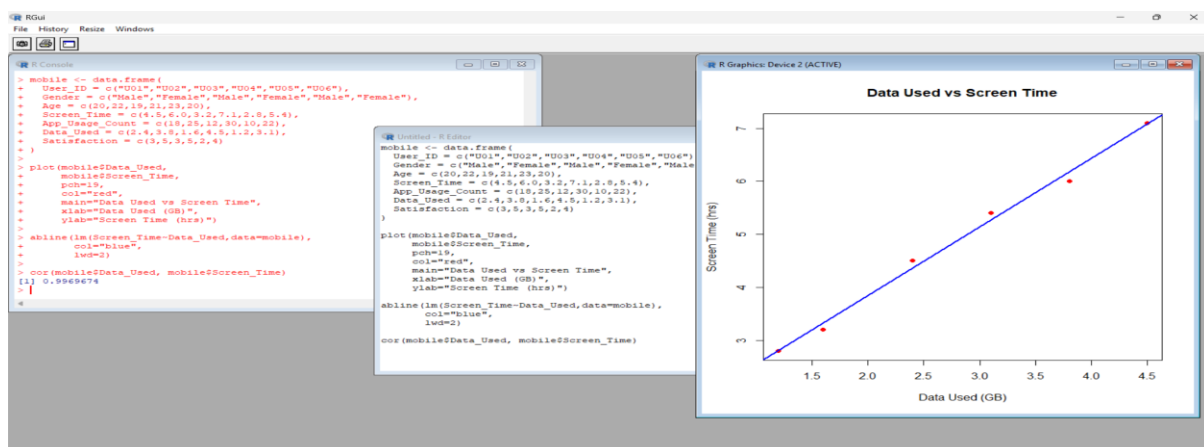
```
User_ID = c("U01","U02","U03","U04","U05","U06"),
Gender = c("Male","Female","Male","Female","Male","Female"),
Age = c(20,22,19,21,23,20),
Screen_Time = c(4.5,6.0,3.2,7.1,2.8,5.4),
App_Usage_Count = c(18,25,12,30,10,22),
Data_Used = c(2.4,3.8,1.6,4.5,1.2,3.1),
Satisfaction = c(3,5,3,5,2,4))

plot(mobile$Data_Used,
     mobile$Screen_Time,
     pch=19,
     col="red",
     main="Data Used vs Screen Time",
     xlab="Data Used (GB)",
     ylab="Screen Time (hrs)")

abline(lm(Screen_Time~Data_Used,data=mobile),
       col="blue",
       lwd=2)

cor(mobile$Data_Used, mobile$Screen_Time)
```

### Output:

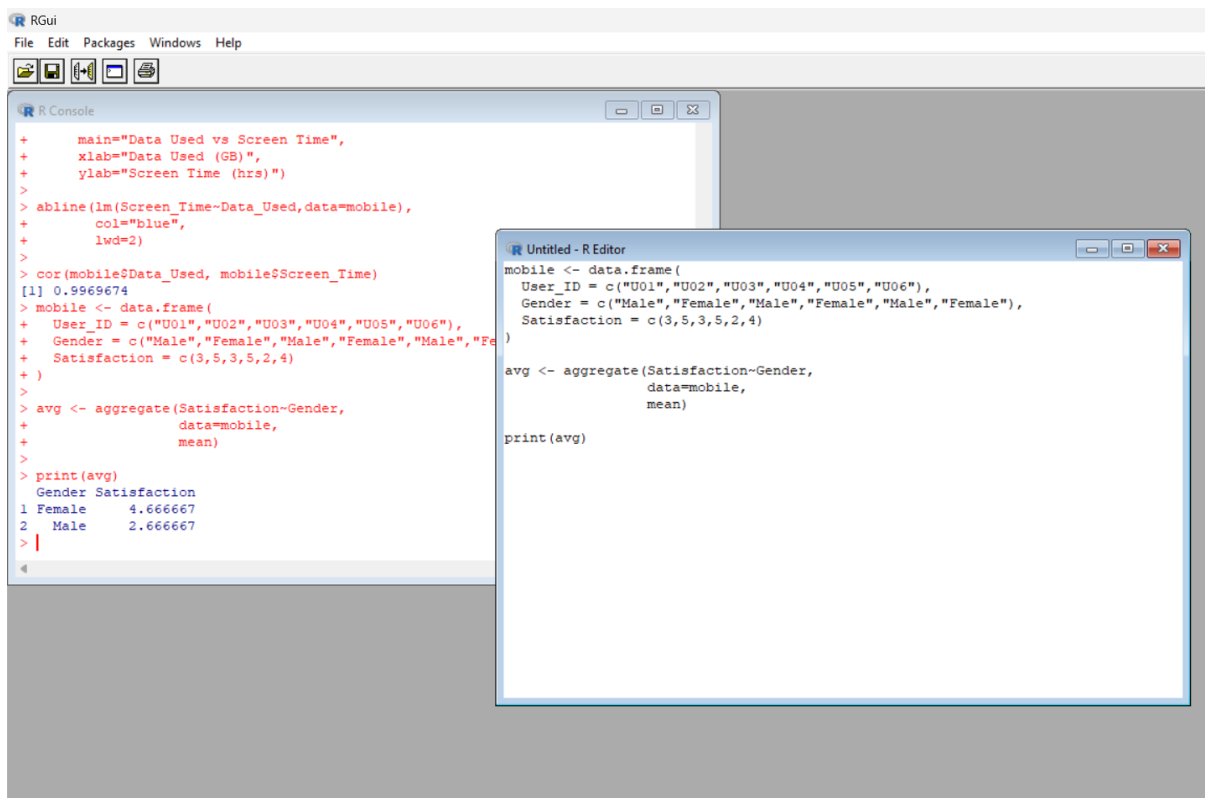


### 30 C. Bar chart for the averages

#### Program:

```
mobile <- data.frame(  
  User_ID = c("U01","U02","U03","U04","U05","U06"),  
  Gender = c("Male","Female","Male","Female","Male","Female"),  
  Satisfaction = c(3,5,3,5,2,4))  
  
avg <- aggregate(Satisfaction~Gender,  
  data=mobile,  
  mean)  
  
print(avg)
```

#### Output:



### Bar Chart for Average Satisfaction

#### Program:

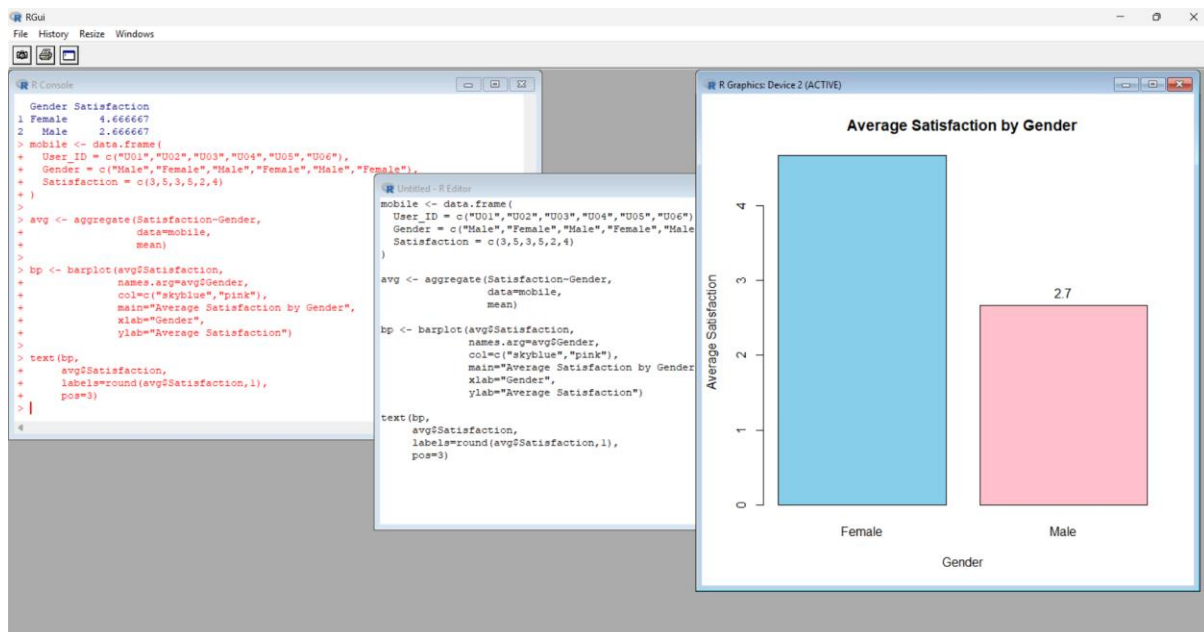
```
mobile <- data.frame(  
  User_ID = c("U01","U02","U03","U04","U05","U06"),  
  Gender = c("Male","Female","Male","Female","Male","Female"),
```

```
Satisfaction = c(3,5,3,5,2,4))
```

```
avg <- aggregate(Satisfaction~Gender,
  data=mobile,
  mean)
```

```
bp <- barplot(avg$Satisfaction,
  names.arg=avg$Gender,
  col=c("skyblue","pink"),
  main="Average Satisfaction by Gender",
  xlab="Gender",
  ylab="Average Satisfaction")
```

```
text(bp,
  avg$Satisfaction,
  labels=round(avg$Satisfaction,1),
  pos=3)
```



### 30 D. Create a bubble chart (Screen Time vs Data Used) a mini dashboard.

#### Program:

```
mobile <- data.frame(
  User_ID = c("U01","U02","U03","U04","U05","U06"),
  Gender = c("Male","Female","Male","Female","Male","Female"),
  Screen_Time = c(4.5,6.0,3.2,7.1,2.8,5.4),
```

```

Data_Used = c(2.4,3.8,1.6,4.5,1.2,3.1),
App_Usage_Count = c(18,25,12,30,10,22))
par(mfrow=c(1,2))
plot(mobile$Screen_Time,
     mobile$Data_Used,
     pch=19,
     cex=mobile$App_Usage_Count/10,
     col="blue",
     main="Screen Time vs Data Used",
     xlab="Screen Time (hrs)",
     ylab="Data Used (GB)")
barplot(mobile$App_Usage_Count,
        names.arg=mobile$User_ID,
        col="lightgreen",
        main="App Usage Count",
        xlab="User",
        ylab="Usage Count")
par(mfrow=c(1,1))

```

## Output:

